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SUPERSEDING
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MILITARY SPECIFICATION

# STRAINERS, SEDIMENT, REFRIGERANT 12

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers noncleanable, screen-type, straight-through strainers for use with refrigerant 12 (dichlorodifluoromethane).

1.2 Classification. The strainers shall be of the following sizes as specified (see 6.2):

| Size | Capacity Tons of refrigeration (T.R.) |
|---|---|
| 1 | 1 to 3 |
| 2 | 2 to 4 |
| 3 | 4 to 10 |

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| BB-F-1421 | - | Fluorocarbon Refrigerants |
| PPP-B-566 | - | Boxes, Folding, Paperboard |
| PPP-B-636 | - | Boxes, Shipping, Fiberboard |
| PPP-B-665 | - | Boxes: Paperboard, Metal Edged and Components |
| PPP-B-676 | - | Boxes, Setup |



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STANDARDS

FEDERAL

FED-STD-356 - Commercial Packaging of Supplies and Equipment

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets
MS 17242 - Strainer, Sediment, Refrigerant-12

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

Underwriters' Laboratories, Inc. (UL)

UL 207(C) - Refrigerant Containing Components

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 1285 Walt Whitman Road, Melville, L.I., NY 11746; or 1655 Scott Boulevard, Santa Clara, CA 95050.)

(Technical society and technical associations specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The strainers delivered under this specification shall be the manufacturer's standard commercial product except for any changes necessary to meet specification requirements.



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3.2 Safety standard. Strainers shall be tested and approved under UL 207(C).

3.2.1 Certification. Prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the strainers he proposes to furnish under this specification meet the applicable requirements of UL 207(C). Acceptable evidence of meeting the requirements of this standard shall be the UL label, an approved UL listing, or a certified test report from a recognized independent testing laboratory acceptable to the Government, indicating that the strainers offered have been tested and conform to the requirements of UL 207(C).

3.3 Materials. The materials shall conform to the applicable specifications and requirements specified in MS 17242. Material not definitely specified shall be of the quality normally used by the supplier in fabricating strainers.

3.4 Design and construction. Design and construction of the strainer shall be as specified herein and in accordance with MS 17242 for use with refrigerant conforming to type 12 of BB-F-1421.

3.4.1 Capacity. The strainer shall have capacities indicated in 1.2 under the conditions specified in MS 17242 (see 4.3.1).

3.5 Marking.

3.5.1 Special markings. The inlet end or the direction of flow shall be clearly etched or embossed on the strainer or on the manufacturer's standard identification decalcomania.

3.5.2 Identification. The manufacturer's name or identifiable trade name shall be clearly stamped, etched, or embossed on a securely attached corrosion-resistant, metal nameplate. In lieu of the metal identification plate, a pressure-sensitive label or waterproofed decalcomania, attached to the shell with an adhesive, may be used. In addition, the strainers shall be marked as required by UL 207(C).

3.6 Finish. The strainers shall have the manufacturer's standard commercial finish. The strainer shall not have any sharp burrs, slivers, or splinters.

3.7 Workmanship. The finished strainer shall not be fractured, split, punctured, dented, or malformed. Soldering shall be complete, clean, and adherent and shall not have any pinholes. Welding and brazing shall be complete, uniform, and properly fused having no burn holes, slag, inclusions, scale or flux deposits, and shall not be cracked, fractured, or undercut.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with the requirements of referenced specifications, standards, and drawings, unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 End product inspection. The lot shall consist of all the strainers of one size offered for inspection at one time. The sample unit for the inspection shall be one strainer.

4.2.2.1 Visual examination. The strainer shall be examined for defects classified in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

| Examine | Defect | Classification Major | Minor |
|---|---|---|---|
| Finish | Finish not manufacturer's standard type | | X |
| | Sharp burrs, slivers, or splinters | X | |
| Construction, workmanship (applicable to all components and assemblies) | Fractured, split, punctured, dented, malformed | X | |
| | Part missing | X | |
| | Not connected or joined or poorly accomplished | X | |



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TABLE I. Classification of defects (cont'd)

| Examine | Defect | Classification Major | Minor |
|---|---|---|---|
| Soldering | Soldering not clean (flux or flux residue not removed); pinholes in solder, not complete, not adherent (when applicable) | | X |
| Welding and brazing | Welding and brazing having slag, inclusions, is undercut, not complete, not properly fused, not uniform, with scale or flux deposits not removed | | X |
| Marking | Identification marking missing, not legible, not specified type, not affixed in the prescribed manner | | X |
| | Inlet end or direction of flow not clearly etched or embossed on strainer or indicated on decalcomania | X | |

4.2.2.2 Dimensional examination. Examination shall be made of the strainer to determine compliance with dimensional requirements. Any dimension not within specified requirements shall be classified a defect. The inspection level shall be S-4 and the AQL shall be 2.5, expressed in terms of defects per hundred units.

* 4.2.3 Testing of the end product. The strainers shall be tested as specified in 4.3.1. The inspection level shall be level II with an AQL of 2.5 expressed in terms of defects per hundred units.

4.2.4 Standards compliance. Proof of compliance with the requirements of UL (see 3.2.1) shall be available to the Government representative.

* 4.2.5 Examination of packaging. An examination shall be made to determine that unit packaging, intermediate packaging, packing and marking as required by



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section 5 of this specification are complied with. Defects shall be as listed in table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

TABLE II. Examination of packaging

| Examine | Defect |
|---|---|
| Marking | Omitted; incorrect; illegible; of improper size, location, sequence, or method of application. |
| Materials | Component missing, damaged, or otherwise defective. |
| Contents | Contents per container is more or less than required. |

4.2.5.1 Inspection of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, inspection for defects in closure, waterproofing, and strapping shall be in accordance with PPP-B-636.

4.2.6 Examination for palletization. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL shall be 6.5 defects per hundred units.

| Examine | Defect |
|---|---|
| Finished dimension | Length, width, or height exceeds specified maximum requirements. |
| Palletization | Not as specified.<br>Pallet pattern not as specified.<br>Interlocking of loads not as specified.<br>Load not bonded with required straps as specified. |
| Weight | Exceeds maximum load limits. |
| Marking | Omitted; incorrect; illegible; of improper size, location, sequence or method of application. |



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4.3 Test.

4.3.1 Capacity test. A capacity test shall be performed to determine compliance with 3.4.1. The strainer shall be connected to a refrigeration system or testing apparatus having a capacity equal to or in excess of the T.R. rating of the strainer to be tested. The strainer shall be tested at the maximum T.R. rating for the size being tested. When the test system is placed in operation, the pressure gages at the inlet and outlet of the strainer shall be observed to insure that the strainer has the capacity and screen area to allow refrigerant 12 liquid to flow through the strainer, at a rate equal to 3.9 pounds per minute miltiplied by the strainer T.R. rating, without creating a pressure drop in excess of 2 pounds per square inch (psi) measured between the strainer inlet and the strainer outlet.

5. PACKAGING

5.1 Preservation - packaging. Preservation - packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each strainer, provided with thread protectors on each connection, shall be unit-packaged in a snug-fitting folding, setup, or metal stayed paperboard box conforming to variety 1, style III or VIII, type G, class 1 of PPP-B-566; type I, variety 1, class A or B, style 2 of PPP-B-676; or style C of PPP-B-665, respectively. Each PPP-B-566 and PPP-B-676 box shall be closed in accordance with the appendix of the applicable box specification. Each PPP-B-665 box shall be closed and the closure secured with a 4-inch strip of gummed paper or pressure-sensitive tape.

* 5.1.2 Level C. The strainers shall be unit-packaged in accordance with FED-STD-356, provided the package and the quantity per package is the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A packing. Strainers, packaged as specified in 5.1, shall be packed in the quantities and arrangement as specified in table III, in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced in accordance with the appendix of PPP-B-636.



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TABLE III. Quantity and arrangement of strainers for shipping container

| Size | Quantity per shipping container (each) | Arrangement of strainers in shipping container | Approximate size of shipping container in inches (see note) |
|---|---|---|---|
| 1 | 72 | 4L by 6W by 3D | 18-1/2L by 14-1/4W by 7-1/8D |
| 2 | 48 | 6L by 2W by 4D | 14-1/4L by 10-1/2W by 9-1/2D |
| 3 | 36 | 2L by 6W by 3D | 14-3/4L by 14-1/2W by 7-1/8D |

NOTE: Approximate dimensions are furnished as a guide only.

5.2.2 Level B packing. Strainers, packaged as specified in 5.1, shall be packed in the quantity and arrangement as specified in table III, in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or type SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification.

* 5.2.3 Level C packing. The strainers, packaged as specified in 5.1, shall be packed in accordance with FED-STD-356, provided the quantity packed per shipping container is the same as that normally used by the contractor for retail distribution.

5.3 Palletization. When specified (see 6.2), strainers of one size only, packed as specified in 5.2, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.



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5.4 Marking. In addition to any special marking required by the contract order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The strainers covered by this specification are intended to withhold foreign matter from a gas or liquid in refrigeration systems.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(e) When palletization is required (see 5.3).

6.3 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER AND TITLE
MIL-S-22833C - Strainers, Sediment, Refrigerant 12

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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